NTA CUET (UG)

NTA CUET (UG) Physics — 29 May 2024

Physics · Full marks 200 · 60 minutes · 50 questions

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`http://localhost:3000/p/cuet-physics-2024-05-29`

Instructions

- Attempt only 40 questions out of the given 50 questions.
- Each question carries 5 marks.
- One mark will be deducted for a wrong answer.

Physics — 5 marks each

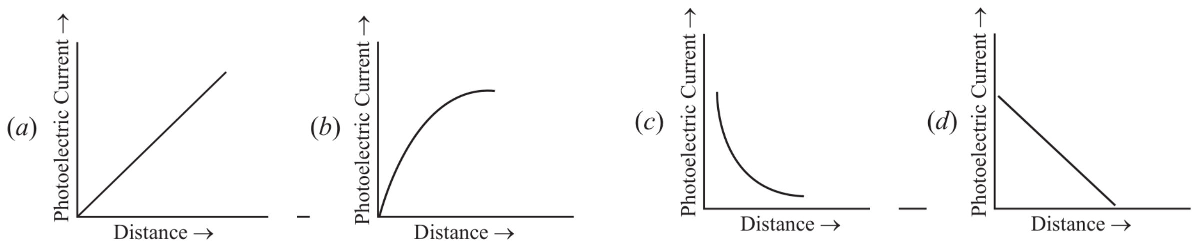
Q1. The kinetic energy of an electron in the ground level of a hydrogen atom is K units. The values of its potential energy and total energy respectively are ______. [5 marks · Easy]

(a) $-2K; -K$
(b) $+2K; -K$
(c) $-K; +2K$
(d) $+K; +2K$

Q2. Two nuclei have mass numbers A and B respectively. The density ratio of the nuclei is ______. [5 marks · Easy]

(a) $A : B$
(b) $\sqrt{A} : \sqrt{B}$
(c) $A^2 : B^2$
(d) $1 : 1$

Q3. A point source causing photoelectric emission from a metallic plate is moved away from the plate. The variation of photoelectric current with distance from the source is correctly represented by the graph ______. [5 marks · Easy]



(a) Graph (a): photoelectric current increases linearly with distance
(b) Graph (b): photoelectric current rises with distance and saturates
(c) Graph (c): photoelectric current falls off sharply with distance (like $1/d^2$), levelling off
(d) Graph (d): photoelectric current decreases linearly with distance

Q4. A proton accelerated through a potential difference V has a de Broglie wavelength λ. On doubling the accelerating potential, de Broglie wavelength of the proton ______. [5 marks · Easy]

(a) remains unchanged
(b) becomes double
(c) becomes four times
(d) decreases

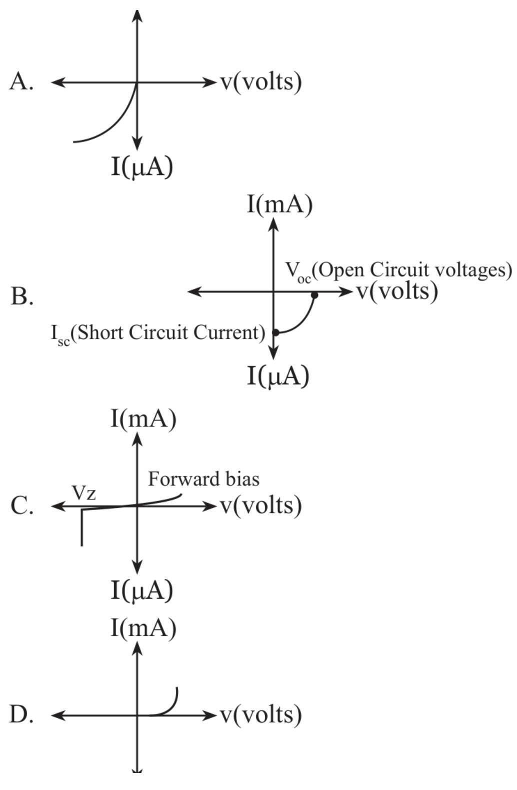
Q5. The shortest wavelengths emitted in hydrogen spectrum corresponding to different spectral series are as under: A. Pfund series, B. Balmer series, C. Brackett series, D. Lyman series. The wavelengths arranged correctly in decreasing order are ______. [5 marks · Medium]

(a) (A), (B), (C), (D)
(b) (A), (C), (B), (D)
(c) (B), (A), (D), (C)
(d) (A), (C), (D), (B)

Q6. Silicon can be doped using one of the following elements as dopant: A. Arsenic, B. Indium, C. Phosphorus, D. Boron. To get n-type semiconductor, the dopants that can be used are ______. [5 marks · Easy]

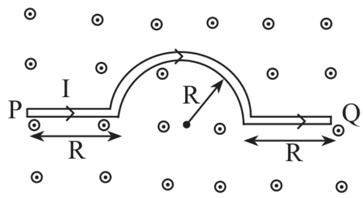
(a) (A) and (C) only
(b) (B) and (C) only
(c) (A), (B), (C) and (D)
(d) (C) and (D) only

Q7. Given below are V versus I graphs for different types of p–n junction diodes marked A, B, C, and D. The correct sequence of graphs corresponding to forward biased p–n junction, Zener diode, photo diode, and solar cell in order is ______. [5 marks · Medium]



(a) (D), (C), (A), (B)
(b) (A), (C), (B), (D)
(c) (B), (A), (D), (C)
(d) (C), (B), (D), (A)

Q8. A wire carrying current I, bent as shown in the figure, is placed in a uniform field B that emerges normally out of the plane of the figure. The force on this wire is ______. [5 marks · Medium]



(a) 4BIR, directed vertically downward
(b) 3BIR, directed vertically upward
(c) $BI(2R + \pi R)$, vertically downward
(d) $2\pi BIR$, from P to Q

Q9. The refractive index of the material of an equilateral prism is $\sqrt{2}$. The angle of minimum deviation of that prism is ______. [5 marks · Medium]

(a) 60°
(b) 75°
(c) 30°
(d) 90°

Q10. The transfer of integral number of ______ is one of the evidence of quantization of electric charge. [5 marks · Easy]

(a) photons
(b) nuclei
(c) electrons
(d) neutrons

Q11. When a slab of insulating material 4 mm thick is introduced between the plates of a parallel plate capacitor of separation 4 mm, it is found that the distance between the plates has to be increased by 3.2 mm to restore the capacity to its original value. The dielectric constant of the material is ______. [5 marks · Medium]

(a) 2
(b) 5
(c) 3
(d) 7

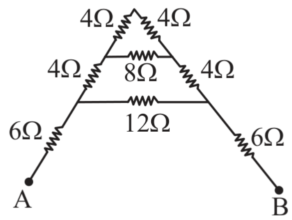
Q12. A copper ball of density 8.0 g/cc and 1 cm in diameter is immersed in oil of density 0.8 g/cc. The charge on the ball if it remains just suspended in oil in an electric field of intensity 600 N/C acting in the upward direction is ______. (Take $g = 10\ m/s^2$) [5 marks · Hard]

(a) 2×10^{-6} C
(b) 2×10^{-5} C
(c) 1×10^{-5} C
(d) 1×10^{-6} C

Q13. A metal wire is subjected to a constant potential difference. When the temperature of the metal wire increases, the drift velocity of the electron in it ______. [5 marks · Easy]

(a) increases, thermal velocity of the electrons decreases
(b) decreases, thermal velocity of the electrons decreases
(c) increases, thermal velocity of the electrons increases
(d) decreases, thermal velocity of the electrons increases

Q14. For the given mixed combination of resistors, calculate the total resistance between points A and B. [5 marks · Medium]



(a) 9 Ω
(b) 18 Ω
(c) 4 Ω
(d) 14 Ω

Q15. A cell of emf 1.1 V and internal resistance 0.5 Ω is connected to a wire of resistance 0.5 Ω. Another cell of the same emf is now connected in series with the intention of increasing the current but the current in the wire remains the same. The internal resistance of the second cell is ______. [5 marks · Medium]

(a) 1 Ω
(b) 2.5 Ω
(c) 1.5 Ω
(d) 2 Ω

Q16. P, Q, R, and S are four wires of resistances 3 Ω, 3 Ω, 3 Ω, and 4 Ω respectively. They are connected to form the four arms of a Wheatstone bridge circuit. The resistance with which S must be shunted in order that the bridge may be balanced is ______. [5 marks · Medium]

(a) 14 Ω
(b) 12 Ω
(c) 15 Ω
(d) 7 Ω

Q17. Magnetic moment of a thin bar magnet is 'M'. If it is bent into a semicircular form, its new magnetic moment will be ______. [5 marks · Medium]

(a) M/π
(b) $M/2$
(c) M
(d) $2M/\pi$

Q18. Ferromagnetic material used in Transformers must have ______. [5 marks · Easy]

(a) Low permeability and High Hysteresis loss
(b) High permeability and Low Hysteresis loss
(c) High permeability and High Hysteresis loss
(d) Low permeability and Low Hysteresis loss

Q19. A conducting ring of radius r is placed in a varying magnetic field perpendicular to the plane of the ring. If the rate at which the magnetic field varies is x, the electric field intensity at any point of the ring is ______. [5 marks · Medium]

(a) rx

(b) $rx/2$
(c) $2rx$
(d) $4\pi r/x$

Q20. A 50 Hz AC current of crest value 1 A flows through the primary of a transformer. If the mutual inductance between the primary and secondary be 0.5 H, the crest voltage induced in the secondary is ______. [5 marks · Medium]

(a) 75 V
(b) 150 V
(c) 100 V
(d) 200 V

Q21. A long solenoid of diameter 0.1 m has 2×10^4 turns per meter. At the center of the solenoid, a coil of 100 turns and radius 0.01 m is placed with its axis coinciding with the solenoid axis. The current in the solenoid reduces at a constant rate to 0 A from 4 A in 0.05 s. If the resistance of the coil is $10\pi^2$ Ω, then the total charge flowing through the coil during this time is ______. [5 marks · Hard]

(a) 16 μC
(b) 32 μC
(c) 16π μC
(d) 32π μC

Q22. Lower half of a convex lens is made opaque. Which of the following statements describes the image of the object placed in front of the lens? A. No change in image. B. Image will show only half of the object. C. Intensity of image gets reduced. [5 marks · Easy]

(a) (A) only
(b) (B) only
(c) (C) only
(d) (B) and (C) only

Q23. Two slits are made 0.1 mm apart and the screen is placed 2 m away. The fringe separation when a light of wavelength 500 nm is used is ______. [5 marks · Easy]

(a) 1 cm
(b) 0.15 cm
(c) 1.5 cm
(d) 0.1 cm

Q24. For an astronomical telescope having an objective lens of focal length 10 m and an eyepiece lens of focal length 10 cm, the telescope's tube length and magnification respectively are ______. [5 marks · Easy]

(a) 20 cm, 1
(b) 1000 cm, 1
(c) 1010 cm, 1
(d) 1010 cm, 100

Q25. According to Bohr's Model: A. The radius of the orbiting electron is directly proportional to 'n'. B. The speed of the orbiting electron is directly proportional to '1/n'. C. The magnitude of the total energy of the orbiting electron is directly proportional to '$1/n^2$'. D. The radius of the orbiting electron is directly proportional to 'n^2'. [5 marks · Medium]

(a) (A), (B) and (C) only
(b) (A), (B) and (D) only
(c) (A), (B), (C) and (D)
(d) (B), (C) and (D) only

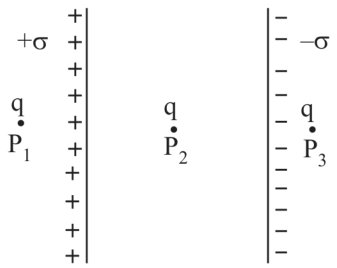
Q26. For a full wave rectifier, if the input frequency is 50 Hz, the output frequency will be ______. [5 marks · Easy]

(a) 50 Hz
(b) 100 Hz
(c) 25 Hz
(d) 0 Hz

Q27. For an electric dipole in a non-uniform electric field with dipole moment parallel to direction of the field, the force (F) and torque (τ) on the dipole respectively are ______. [5 marks · Medium]

(a) $F = 0, \tau = 0$
(b) $F \neq 0, \tau = 0$
(c) $F = 0, \tau \neq 0$
(d) $F \neq 0, \tau \neq 0$

Q28. Two large plane parallel sheets shown in the figure have equal but opposite surface charge densities $+\sigma$ and $-\sigma$. A point charge q placed at points P1, P2, and P3 experiences forces F1, F2, and F3 respectively. Then: [5 marks · Medium]



(a) $F_1 = 0, F_2 = 0, F_3 = 0$
(b) $F_1 = 0, F_2 \neq 0, F_3 = 0$
(c) $F_1 \neq 0, F_2 \neq 0, F_3 \neq 0$
(d) $F_1 = 0, F_3 \neq 0, F_2 = 0$

Q29. Two charged metallic spheres with radii R_1 and R_2 are brought in contact and then separated. The ratio of final charges Q_1 and Q_2 on the two spheres respectively will be ______. [5 marks · Easy]

(a) $Q_1/Q_2 = R_2/R_1$
(b) $Q_1/Q_2 < R_1/R_2$
(c) $Q_1/Q_2 > R_1/R_2$
(d) $Q_1/Q_2 = R_1/R_2$

Q30. Two charged particles, placed at a distance d apart in vacuum, exert a force F on each other. Now, each of the charges is doubled. To keep the force unchanged, the distance between the charges should be changed to ______. [5 marks · Easy]

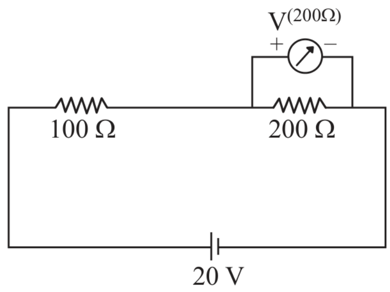
(a) 4d
(b) 2d

(c) d
(d) d/2

Q31. Two parallel plate capacitors of capacitances 2 μF and 3 μF are joined in series and the combination is connected to a battery of V volts. The values of potential across the two capacitors V_1 and V_2 and energy stored in the two capacitors U_1 and U_2 respectively are related as ______. [5 marks · Medium]

(a) $V_1/V_2 = U_1/U_2 = 3/2$
(b) $V_1/V_2 = 2/3$ and $U_1/U_2 = 3/2$
(c) $V_1/V_2 = 3/2$ and $U_1/U_2 = 2/3$
(d) $V_1/V_2 = 2/3$ and $U_1/U_2 = 2/3$

Q32. Two resistances of 100 Ω and 200 Ω are connected in series across a 20 V battery as shown in the figure below. The reading in a 200 Ω voltmeter connected across the 200 Ω resistance is ______. [5 marks · Medium]



(a) 4 V
(b) 20/3 V
(c) 10 V
(d) 16 V

Q33. The current through a 4/3 Ω external resistance connected to a parallel combination of two cells of 2 V and 1 V emf and internal resistances of 1 Ω and 2 Ω respectively is ______. [5 marks · Medium]

(a) 1 A
(b) 2/3 A
(c) 8/9 A
(d) 5/6 A

Q34. A metallic wire of uniform area of cross-section has a resistance R, resistivity ρ, and power rating P at V volts. The wire is uniformly stretched to reduce the radius to half the original radius. The values of resistance, resistivity, and power rating at V volts are now denoted by R′, ρ', and P′ respectively. The corresponding values are correctly related as ______. [5 marks · Medium]

(a) $\rho' = 2\rho$, $R' = 2R$, $P' = 2P$
(b) $\rho' = (1/2)\rho$, $R' = (1/2)R$, $P' = (1/2)P$
(c) $\rho' = \rho$, $R' = 16R$, $P' = (1/16)P$
(d) $\rho' = \rho$, $R' = (1/16)R$, $P' = 16P$

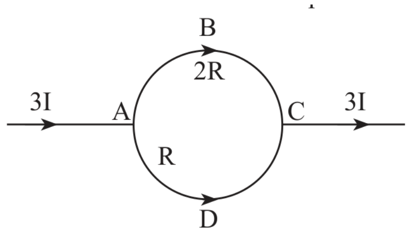
Q35. Three magnetic materials are listed below: A. Paramagnetics, B. Diamagnetics, C. Ferromagnetics. Choose the correct order of the materials in increasing order of magnetic susceptibility. [5 marks · Easy]

(a) (A), (B), (C)
(b) (C), (A), (B)
(c) (B), (A), (C)
(d) (B), (C), (A)

Q36. Two infinitely long straight parallel conductors carrying currents I_1 and I_2 are held at a distance d apart in vacuum. The force F on a length L of one of the conductors due to the other is ______. [5 marks · Easy]

(a) proportional to L but independent of $I_1 \times I_2$
(b) proportional to $I_1 \times I_2$ but independent of length L
(c) proportional to $I_1 \times I_2 \times L$
(d) proportional to $L/(I_1 \times I_2)$

Q37. In the circuit shown below, a current 3I enters at A. The semicircular parts ABC and ADC have equal radii 'r' but resistances 2R and R respectively. The magnetic field at the center of the circular loop ABCD is ______. [5 marks · Hard]



(a) $(\mu_0 I)/(4r)$ out of the plane
(b) $(\mu_0 I)/(4r)$ into the plane
(c) $(\mu_0 3I)/(4r)$ out of the plane
(d) $(\mu_0 3I)/(4r)$ into the plane

Q38. A square loop with each side 1 cm, carrying a current of 10 A, is placed in a magnetic field of 0.2 T. The direction of the magnetic field is parallel to the plane of the loop. The torque experienced by the loop is ______. [5 marks · Medium]

(a) zero
(b) 2×10^{-4} Nm
(c) 2×10^{-2} Nm
(d) 2 Nm

Q39. In an AC circuit, the current leads the voltage by $\pi/2$. The circuit is ______. [5 marks · Easy]

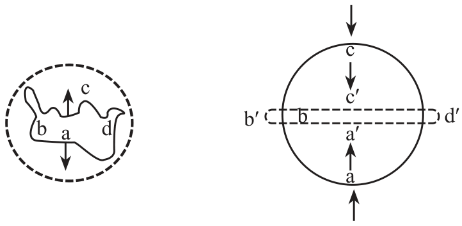
(a) purely resistive
(b) should have circuit elements with resistance equal to reactance
(c) purely inductive
(d) purely capacitive

Q40. In a pair of adjacent coils, for a change of current in one of the coils from 0 A to 10 A in 0.25 s, the magnetic flux in the adjacent coil changes by 15 Wb. The mutual inductance of the coils is ______. [5 marks · Easy]

(a) 120 H
(b) 12 H
(c) 1.5 H

(d) 0.75 H

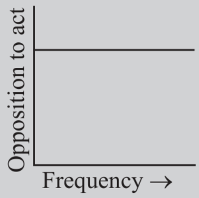
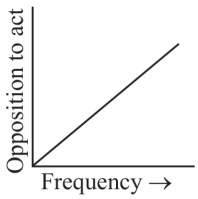
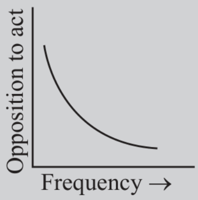
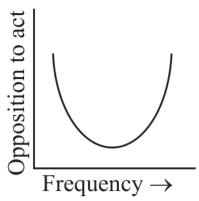
Q41. A wire of irregular shape in figure (a) and a circular loop of wire in figure (b) are placed in different uniform magnetic fields as shown in the figures below. In figure (a), the magnetic field is perpendicular into the plane. In figure (b), the magnetic field is perpendicular out of the plane. The wire in figure (a) is turning into a circular loop and that in figure (b) into a narrow straight wire. The direction of induced current will be ______. [5 marks · Medium]



(a) (b)

(a) clockwise in both (a) and (b)
(b) anticlockwise in both (a) and (b)
(c) clockwise in (a) and anticlockwise in (b)
(d) anticlockwise in (a) and clockwise in (b)

Q42. Match List–I has four graphs showing variation of opposition to flow of AC versus frequency with circuit characteristic in List–II. List–II: (I) Impedance, (II) Capacitive reactance, (III) Inductive reactance, (IV) Resistance. [5 marks · Medium]

List–I	List–II
(A) 	(I) Impedance
(B) 	(II) Capacitive reactance
(C) 	(III) Inductive reactance
(D) 	(IV) Resistance

(a) (A) – (I), (B) – (III), (C) – (II), (D) – (IV)
(b) (A) – (IV), (B) – (III), (C) – (II), (D) – (I)
(c) (A) – (I), (B) – (II), (C) – (IV), (D) – (III)
(d) (A) – (III), (B) – (IV), (C) – (I), (D) – (II)

Q43. In an electromagnetic wave, the ratio of energy densities of electric and magnetic fields is ______. [5 marks · Easy]

(a) 1 : 1
(b) 1 : c
(c) c : 1
(d) 1 : c^2

Q44. Of the following, the correct arrangement of electromagnetic spectrum in decreasing order of wavelength is ______. [5 marks · Easy]

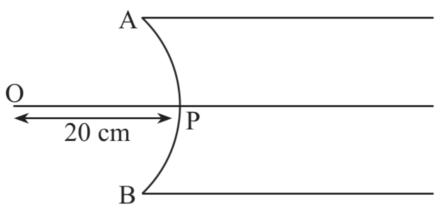
(a) Radio waves, X-rays, Infrared waves, microwaves, visible waves
(b) Infrared waves, microwaves, Radio waves, X-rays, visible waves
(c) Radio waves, microwaves, Infrared waves, visible waves, X-rays
(d) X-rays, visible waves, Infrared waves, microwaves, Radio waves

Q45. Match Electromagnetic waves listed in List–I with Production method/device in List–II. List–I: (A) Microwaves, (B) Infrared, (C) X-rays, (D) Radio waves. List–II: (I) LC oscillator, (II) Magnetron, (III) Vibration of atoms/molecules, (IV) Bombarding large atomic number metal target with moving electrons. [5 marks · Easy]

List–I	List–II
(A) Microwaves	(I) LC oscillator
(B) Infrared	(II) Magnetron
(C) X-rays	(III) Vibration of atoms/molecules
(D) Radio waves	(IV) Bombarding large atomic number metal target with moving electrons

(a) (A) – (I), (B) – (II), (C) – (III), (D) – (IV)
(b) (A) – (II), (B) – (III), (C) – (IV), (D) – (I)
(c) (A) – (II), (B) – (I), (C) – (IV), (D) – (III)
(d) (A) – (III), (B) – (IV), (C) – (I), (D) – (II)

Q46. In the figure given below, APB is a curved surface of radius of curvature 10 cm separating air and a transparent material ($\mu = 4/3$). A point object O is placed in air on the principal axis of the surface 20 cm from P. The distance of the image of O from P will be ______. [5 marks · Medium]



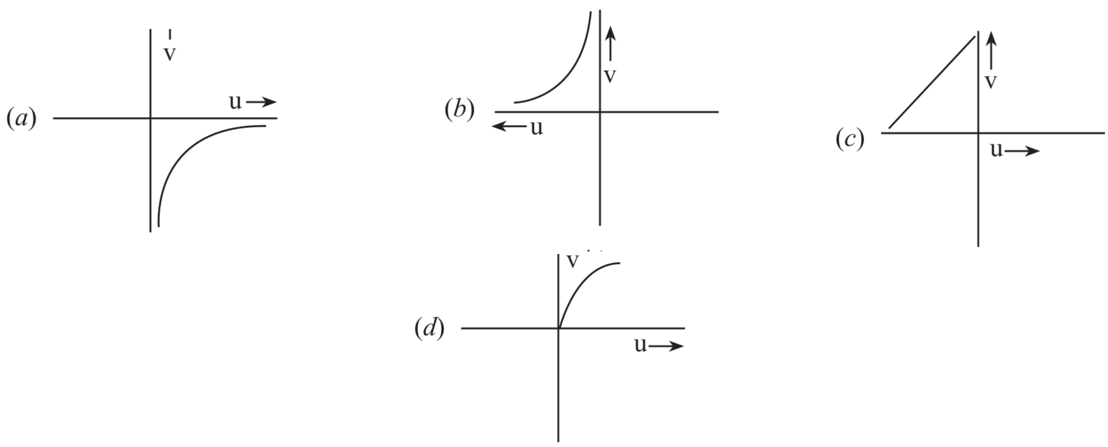
(a) 16 cm left of P in air

(b) 16 cm right of P in water
(c) 20 cm right of P in water
(d) 20 cm left of P in air

Q47. For fixed values of radii of curvature of lens, power of the lens will be ______. [5 marks · Easy]

(a) $P \propto (\mu - 1)$
(b) $P \propto \mu^2$
(c) $P \propto 1/\mu$
(d) $P \propto \mu^{-2}$

Q48. The graph correctly representing the variation of image distance 'v' for a convex lens of focal length 'f' versus object distance 'u' is ______. [5 marks · Medium]



(a) Graph (a): hyperbolic branch in the fourth quadrant (positive u, negative v)
(b) Graph (b): hyperbolic branch in the second quadrant (negative u, positive v)
(c) Graph (c): straight line with positive slope through the origin
(d) Graph (d): curve rising from the origin and saturating for positive u

Q49. Using light from a monochromatic source to study diffraction in a single slit of width 0.1 mm, the linear width of central maxima is measured to be 5 mm on a screen held 50 cm away. The wavelength of light used is ______. [5 marks · Medium]

(a) 2.5×10^{-7} m
(b) 4×10^{-7} m
(c) 5×10^{-7} m
(d) 7.5×10^{-7} m

Q50. Radiation of frequency $2\nu_0$ is incident on a metal with threshold frequency ν_0. The correct statement of the following is _____. [5 marks · Easy]

(a) No photoelectrons will be emitted
(b) All photoelectrons emitted will have kinetic energy equal to $h\nu_0$
(c) Maximum kinetic energy of photoelectrons emitted can be $h\nu_0$
(d) Maximum kinetic energy of photoelectrons emitted will be $2h\nu_0$

Answer Key

1	(a)	**2**	(d)	**3**	(c)	**4**	(d)	**5**	(b)
6	(a)	**7**	(a)	**8**	(a)	**9**	(c)	**10**	(c)
11	(b)	**12**	(b)	**13**	(d)	**14**	(b)	**15**	(a)
16	(b)	**17**	(d)	**18**	(b)	**19**	(b)	**20**	(c)
21	(b)	**22**	(c)	**23**	(a)	**24**	(d)	**25**	(d)
26	(b)	**27**	(b)	**28**	(b)	**29**	(d)	**30**	(b)
31	(a)	**32**	(c)	**33**	(d)	**34**	(c)	**35**	(c)
36	(c)	**37**	(a)	**38**	(b)	**39**	(d)	**40**	(c)
41	(b)	**42**	(b)	**43**	(a)	**44**	(c)	**45**	(b)
46	(a)	**47**	(a)	**48**	(b)	**49**	(c)	**50**	(c)

Solutions

Transcribed from the paper's own answer key and explanations.

Q1.

Answer: (a) $-2K; -K$

- **Recall Bohr-orbit energy relations:** For an electron in the ground state of a hydrogen atom, Kinetic Energy (KE) = K.
- **Potential energy:** $PE = -2 \times KE = -2K$.
- **Total energy:** $TE = KE + PE = K - 2K = -K$.

In the Bohr model the potential energy is always minus twice the kinetic energy, so with KE = K the potential energy is -2K and the total energy is K - 2K = -K.

Q2.

Answer: (d) 1 : 1

- **Key fact:** The density of a nucleus is independent of its mass number.
- **Conclude:** Therefore, the density ratio of any two nuclei is always 1 : 1.

Nuclear density is a constant (~2.3 × 10^17 kg/m^3) independent of mass number, so any two nuclei have a density ratio of 1 : 1.

Q3.

Answer: (c) Graph (c): photoelectric current falls off sharply with distance (like $1/d^2$), levelling off

- **Intensity falls with distance:** As the point source is moved away from the metallic plate, the intensity of light falling on the plate decreases with distance.
- **Current follows intensity:** This decrease in intensity reduces the photoelectric current; the correct graph showing this inverse relationship is graph (c).

Intensity from a point source falls as $1/d^2$, and photoelectric current is proportional to intensity, so the current decays like $1/d^2$ with distance — the steeply falling curve (c).

Q4.

Answer: (d) decreases

- **de Broglie relation:** $\lambda = h/\sqrt{2meV}$, where h is Planck's constant, m the mass, e the charge, and V the potential difference.
- **Double the potential:** $\lambda' = h/\sqrt{2me(2V)} = \lambda/\sqrt{2}$.
- **Conclude:** Thus the de Broglie wavelength decreases.

Since $\lambda \propto 1/\sqrt{V}$, doubling the accelerating potential reduces the wavelength by a factor of $\sqrt{2}$ — it decreases.

Q5.

Answer: (b) (A), (C), (B), (D)

- **Series limits:** Lyman series (n = 1, n = ∞) has the shortest wavelength; Balmer (n = 2), Brackett (n = 4), Pfund (n = 5, longest wavelength).
- **Order:** Decreasing wavelength order: Pfund (A), Brackett (C), Balmer (B), Lyman (D).

The series-limit wavelength grows with the lower level n, so Pfund (n=5) > Brackett (n=4) > Balmer (n=2) > Lyman (n=1) — order (A), (C), (B), (D).

Q6.

Answer: (a) (A) and (C) only

- **n-type needs pentavalent dopants:** For n-type semiconductors, elements with five valence electrons (pentavalent impurities) are used as dopants.
- **Identify pentavalent elements:** Arsenic (A) and Phosphorus (C) are pentavalent impurities, making them suitable for creating n-type semiconductors.

n-type doping requires pentavalent atoms that donate an extra electron; of the list, only arsenic and phosphorus are pentavalent (indium and boron are trivalent, giving p-type).

Q7.

Answer: (a) (D), (C), (A), (B)

- **Identify each graph:** (A) is a photo diode — current proportional to light intensity with a reverse characteristic. (B) is a solar cell — shows short-circuit current (I_{sc}) and open-circuit voltage (V_{oc}).
- **Remaining graphs:** (C) is a Zener diode — sharp breakdown in reverse bias. (D) is a forward biased p–n junction diode — exponential increase in current with voltage.
- **Sequence:** Forward biased p–n junction, Zener, photo diode, solar cell → (D), (C), (A), (B).

Matching each characteristic curve to its device (exponential forward rise = ordinary diode, reverse breakdown = Zener, reverse light-dependent current = photodiode, fourth-quadrant curve with Voc and Isc = solar cell) gives the sequence (D), (C), (A), (B).

Q8.

Answer: (a) 4BIR, directed vertically downward

- **Force law:** The force on a current-carrying wire in a magnetic field is $F = I(L \times B)$, where L is the length of the wire in the direction of the field.
- **Straight segments:** For the straight segments the force will be $2 \times BIR$ downward.
- **Semicircular segment:** For the semicircular segment of radius R the force is directed downward with $F = I(2R)B$, since the effective length of the semicircle is 2R.
- **Total:** Thus the force on the wire is 4BIR, vertically downward.

Each part contributes a force B·I·(effective straight length): 2R from the two straight pieces and 2R (the diameter) from the semicircle, giving 4BIR directed vertically downward.

Q9.

Answer: (c) 30°

- **Minimum deviation formula:** $n = \sin[(A+\delta_{min})/2]/\sin(A/2)$ with $A = 60°$ for an equilateral prism.
- **Substitute:** $\sqrt{2} = \sin[(60° + \delta_{min})/2]/0.5$, so $\sin[(60° + \delta_{min})/2] = 1/\sqrt{2}$.
- **Solve:** $(60° + \delta_{min})/2 = 45°$, hence $60° + \delta_{min} = 90°$ and $\delta_{min} = 30°$.

Applying the prism formula $n = \sin((A+\delta min)/2)/\sin(A/2)$ with $n = \sqrt{2}$ and $A = 60°$ forces $(60°+\delta min)/2 = 45°$, so the minimum deviation is 30°.

Q10.

Answer: (c) electrons

- **Quantization of charge:** Quantization of electric charge means charge exists in discrete packets rather than in a continuous manner.
- **Evidence:** This is evidenced by the transfer of integral numbers of electrons, as electrons are the fundamental charge carriers.

Charge changes only by whole multiples of e because charging happens by transfer of an integral number of electrons, the fundamental charge carriers.

Q11.

Answer: (b) 5

- **Capacitance with dielectric:** $C = \kappa\varepsilon_0 A/d$; with a slab of thickness t the effective separation is $d_{eff} = d - t + t/\kappa$.
- **Apply restore condition:** The distance is increased by 3.2 mm to restore the capacity: $3.2 + 4/\kappa = 4$ (with $d - t = 3.2$ and $t = 4$).
- **Solve:** $\kappa = 5$.

Inserting a slab of thickness t reduces the effective separation to d - t + t/κ; restoring the original capacitance after moving the plates 3.2 mm apart requires 3.2 + 4/κ = 4, so κ = 5.

Q12.

Answer: (b) 2×10^{-5} C (per the Answer Key; the paper's own explanation computes 6.288×10^{-5} C and states 'no option match')

- **Set up quantities:** ρ_c = 8.0 g/cc = 8000 kg/m^3, ρ_o = 0.8 g/cc = 800 kg/m^3; volume $V = (4/3)\pi(0.005)^3 = 5.24 \times 10^{-7}$ m^3; mass $m = \rho_c V = 4.192 \times 10^{-3}$ kg.
- **Forces:** Buoyant force $F_b = \rho_o V g = 4.192 \times 10^{-4}$ N (printed as 4.192×10^{-3} N); electric force $F_e = qE$ with E = 600 N/C.
- **Balance:** $qE + \frac{4}{3}\pi r^3 \rho_o g = \frac{4}{3}\pi r^3 \rho_{cu} g$, so $q = \frac{(4/3)\times 3.14\times(0.005)^3\times 7.2\times 10^3\times 10}{600}$.
- **Result (as printed):** $q = 6.288 \times 10^{-5}$ C — 'no option match.' The Answer Key nevertheless marks (b).

Balancing weight against buoyancy plus the upward electric force gives q = V(ρcu - ρoil)g / E; the paper's own arithmetic yields 6.288 × 10^{-5} C, which matches no option, though the key marks (b) 2 × 10^{-5} C.

Q13.

Answer: (d) decreases, thermal velocity of the electrons increases

- **Effect of temperature on collisions:** When the temperature increases, the lattice ions vibrate more vigorously, increasing the rate of collisions between electrons and ions.
- **Consequences:** The increased collision rate reduces the drift velocity (v_d); however, the thermal velocity (v_t) of the electrons increases with temperature.

Higher temperature means more frequent electron–ion collisions (shorter relaxation time), lowering drift velocity, while random thermal speeds grow with temperature.

Q14.

Answer: (b) 18 Ω

- **Top branch:** Combine the two 4 Ω resistors in series at the top: $R_{s1} = 4 + 4 = 8$ Ω; this 8 Ω in parallel with the 8 Ω resistor gives $R_{p1} = 4$ Ω.
- **Middle reduction:** Combine that 4 Ω with the two side 4 Ω resistors in series: $R_{s2} = 4+4+4 = 12$ Ω; this 12 Ω in parallel with the 12 Ω resistor gives $R_{p2} = 6$ Ω.
- **Final series sum:** Combine the 6 Ω with the two 6 Ω legs in series: $R_{s3} = 6 + 6 + 6 = 18$ Ω.

Repeated series–parallel reduction of the ladder (8‖8 → 4, then 4+4+4 = 12, 12‖12 → 6, then 6+6+6) gives 18 Ω between A and B.

Q15.

Answer: (a) 1 Ω

- **Initial current:** $I = E_1/(r_1 + R) = 1.1/(0.5 + 0.5) = 1.1$ A.
- **With second cell in series:** Total emf $= E_1 + E_2 = 2.2$ V and total internal resistance $= r_1 + r_2$; current unchanged: $1.1 = 2.2/(0.5 + r_2 + 0.5)$.
- **Solve:** $1.1(1 + r_2) = 2.2 \Rightarrow r_2 = 1\ \Omega$.

Doubling the emf while keeping the same 1.1 A current means the total resistance must also double from 1 Ω to 2 Ω, so the second cell's internal resistance is 1 Ω.

Q16.

Answer: (b) 12 Ω

- **Balance condition:** In a balanced Wheatstone bridge $P/Q = R/(S\|x)$, where $S\|x$ is S in parallel with the shunt x.
- **Substitute:** With P = Q = R = 3 Ω and S = 4 Ω: $3/3 = 3/(4\|x)$, so $4\|x = 3\ \Omega$.
- **Solve for x:** $4x/(4 + x) = 3 \Rightarrow 4x = 12 + 3x \Rightarrow x = 12\ \Omega$.

Balance requires the fourth arm to equal 3 Ω; shunting the 4 Ω wire with x must give 4x/(4+x) = 3, which solves to x = 12 Ω.

Q17.

Answer: (d) $2M/\pi$

- **Original moment:** $M = m \times l$, the product of pole strength m and length l.
- **Bend into semicircle:** The length becomes the semicircular arc: $l = \pi r$, so $M = m\pi r$; the new pole separation is the diameter $2r$ (original length $l = \pi r$, so $2r = 2l/\pi$).
- **New moment:** $M' = m \times 2r = \dfrac{2}{\pi}M$.

Bending shortens the effective pole-to-pole distance from the arc length πr to the diameter 2r, scaling the moment by $2/\pi$: $M' = 2M/\pi$.

Q18.

Answer: (b) High permeability and Low Hysteresis loss

- **Permeability requirement:** High permeability allows the magnetic flux to pass through easily and ensures efficient operation.
- **Hysteresis requirement:** Low hysteresis loss minimizes energy losses during the magnetization and demagnetization cycles.

A transformer core should channel flux easily (high permeability) and waste little energy per magnetization cycle (low hysteresis loss).

Q19.

Answer: (b) $rx/2$

- **Faraday's law:** $\varepsilon = -d\Phi/dt$ with $\Phi = B\pi r^2$, so $\varepsilon = -\pi r^2(dB/dt) = -\pi r^2 x$.
- **Relate emf to E:** $\varepsilon = E \times 2\pi r$.
- **Solve:** $E = \pi r^2 x/2\pi r = rx/2$.

The induced emf $\pi r^2 x$ is distributed around the circumference $2\pi r$, so the induced electric field is E = rx/2.

Q20.

Answer: (c) 100 V (per the Answer Key; source anomaly — the paper's own explanation computes ≈157 V and concludes 'the closest option is 150 V', i.e. option (b))

- **Induced voltage formula:** $V_S = M(dI_p/dt)$; for $I_p = I_0 \sin(\omega t)$ the maximum rate of change is $I_0\omega$.

- **Substitute:** $I_0 = 1$ A, $f = 50$ Hz, $\omega = 100\pi$ rad/s, so $dI_p/dt = 100\pi$ A/s.
- **Compute:** $V_s = 0.5 \times 100\pi = 50\pi \approx 157$ V; the explanation states the closest option is 150 V, yet the key marks (c) 100 V.

The crest secondary voltage is $M \cdot \omega \cdot I_0 = 0.5 \times 100\pi \times 1 \approx 157$ V; the printed solution rounds to the closest option 150 V, but the printed answer key marks (c) 100 V — an internal inconsistency in the source.

Q21.

Answer: (b) 32 μC

- **Flux through the coil:** $\Phi = B \times A = \mu_0 n I \pi r^2$ per turn; given $\mu_0 = 4\pi \times 10^{-7}$ Tm/A, $n = 2 \times 10^4$ /m, $I = 4$ A, $r = 0.01$ m, $N = 100$.
- **Rate of change:** $dI/dt = (4 - 0)/0.05 = 80$ A/s, so $d\Phi/dt = \mu_0 n (dI/dt) \pi r^2 = 6.4\pi^2 \times 10^{-3}$ Wb/s.
- **Induced emf:** $\varepsilon = -N(d\Phi/dt) = -100 \times 6.4\pi^2 \times 10^{-3} = -0.64\pi^2$ V.
- **Charge:** $Q = \varepsilon \Delta t / R = 0.64\pi^2 \times 0.05/(10\pi^2)$; the explanation evaluates this to 32 μC (its intermediate exponent line is misprinted).

The charge through the coil is $Q = N\Delta\Phi/R = N \cdot \mu_0 n \cdot \Delta I \cdot \pi r^2/R = 100 \times 4\pi \times 10^{-7} \times 2 \times 10^4 \times 4 \times \pi \times 10^{-4}$ / $(10\pi^2) = 3.2 \times 10^{-5}$ C = 32 μC.

Q22.

Answer: (c) (C) only

- **Full image still forms:** When the lower half of a convex lens is made opaque, the lens still forms the complete image of the object.
- **Intensity drops:** Only half the lens converges light, so the amount of light passing through is reduced — the image intensity decreases, but the full object is still visible.

Every part of a lens forms the whole image, so blocking half the lens only halves the light gathered: the complete image forms with reduced intensity.

Q23.

Answer: (a) 1 cm

- **Fringe-width formula:** $\beta = \lambda D/d$ with $\lambda = 500 \times 10^{-9}$ m, $D = 2$ m, $d = 0.1 \times 10^{-3}$ m.
- **Substitute and evaluate:** $\beta = (500 \times 10^{-9} \times 2)/(0.1 \times 10^{-3}) = 10^{-2}$ m $= 1$ cm.

Young's double-slit fringe width $\beta = \lambda D/d = (500 \text{ nm} \times 2 \text{ m})/(0.1 \text{ mm}) = 10$ mm = 1 cm.

Q24.

Answer: (d) 1010 cm, 100

- **Tube length:** $L = f_o + f_e = 1000 + 10 = 1010$ cm.
- **Magnification:** $M = f_o/f_e = 1000/10 = 100$.

In normal adjustment the tube length is fo + fe = 1010 cm and the magnification is fo/fe = 100.

Q25.

Answer: (d) (B), (C) and (D) only

- **Radius:** $r \propto n^2$, so statement (D) is correct and (A) — radius proportional to n — is incorrect.
- **Speed and energy:** $v \propto 1/n$ (statement B correct) and $|E| \propto 1/n^2$ (statement C correct).
- **Conclude:** Correct statements: (B), (C) and (D).

Bohr's model gives $r \propto n^2$, $v \propto 1/n$ and $|E| \propto 1/n^2$, so B, C and D hold while A ($r \propto n$) is false.

Q26.

Answer: (b) 100 Hz

- **Full-wave action:** A full wave rectifier inverts every half cycle of the input AC signal, thereby doubling the frequency of the output signal.
- **Compute:** Output frequency = 2×50 Hz = 100 Hz.

A full-wave rectifier flips the negative half-cycles, producing two output humps per input cycle — the ripple frequency doubles to 100 Hz.

Q27.

Answer: (b) $F \neq 0, \tau = 0$

- **Force in non-uniform field:** The force on the dipole is non-zero ($F \neq 0$) because the field is non-uniform, causing a net force on the dipole.
- **Torque when aligned:** The torque is zero ($\tau = 0$) because the dipole is aligned with the field and there is no tendency to rotate.

A field gradient pulls the two charges unequally (net force), but with p parallel to E the torque p $\times$ E vanishes.

Q28.

Answer: (b) $F_1 = 0$, $F_2 \neq 0$, $F_3 = 0$

- **Outside the sheets:** At P_1 (and likewise P_3) the fields of the two oppositely charged sheets cancel, so the point charge experiences no force: $F_1 = 0$, $F_3 = 0$.
- **Between the sheets:** At P_2 the fields due to both sheets are in the same direction and do not cancel, so $F_2 \neq 0$.

For a pair of equal-and-opposite charged sheets the field is confined to the gap (σ/ε_0 between, zero outside), so only the charge at P2 feels a force.

Q29.

Answer: (d) $Q_1/Q_2 = R_1/R_2$

- **Contact equalizes potential:** In contact the spheres share charge until they reach the same potential; $V = Q/R$ for a sphere.
- **Equate potentials:** $V_1 = V_2 \Rightarrow Q_1/R_1 = Q_2/R_2$, hence $Q_1/Q_2 = R_1/R_2$.

Touching conductors reach a common potential Q/R, so the final charges divide in proportion to the radii.

Q30.

Answer: (b) 2d

- **Coulomb's law:** $F = kq_1q_2/d^2$; doubling each charge gives $F' = k(4q_1q_2)/d'^2$.
- **Keep force unchanged:** $k(4q_1q_2)/d'^2 = k(q_1q_2)/d^2 \Rightarrow d'^2 = 4d^2 \Rightarrow d' = 2d$.

Doubling both charges multiplies the numerator of Coulomb's law by 4, so the separation must double ($d^2 \to 4d^2$) to keep F the same.

Q31.

Answer: (a) $V_1/V_2 = U_1/U_2 = 3/2$

- **Series → same charge:** In series the charge Q on each capacitor is the same; $V_1 = Q/C_1 = Q/2$, $V_2 = Q/C_2 = Q/3$, so $V_1/V_2 = 3/2$.
- **Energy ratio:** $U = Q^2/2C$, so $U_1 = Q^2/4$, $U_2 = Q^2/6$ and $U_1/U_2 = 6/4 = 3/2$.

With equal series charge, both voltage and energy scale as 1/C, so $V_1/V_2 = U_1/U_2 = C_2/C_1 = 3/2$.

Q32.

Answer: (c) 10 V

- **Voltmeter loads the circuit:** The 200 Ω voltmeter in parallel with the 200 Ω resistor gives an effective $R_2 = 100\ \Omega$.
- **Total resistance and current:** $R_{total} = 100 + 100 = 200\ \Omega$; $I = 20/200 = 1/10$ A.
- **Voltmeter reading:** $V = I \times 100\ \Omega = 10$ V (printed as $V_{200} = 1/10$ A $\times$ 200 Ω across the combination position).

The non-ideal voltmeter halves the 200 Ω branch to 100 Ω; the 20 V then divides equally between two 100 Ω sections, so the meter reads 10 V.

Q33.

Answer: (d) 5/6 A

- **Equivalent emf:** $E_{eq} = (E_1/r_1+E_2/r_2)/(1/r_1+1/r_2) = (2/1+1/2)/(1/1+1/2) = 2.5/1.5 = 5/3$ V.
- **Equivalent internal resistance:** $1/r_{eq} = 1/1 + 1/2 = 1.5$, so $r_{eq} = 2/3\ \Omega$.
- **Current:** $R_{total} = 2/3 + 4/3 = 2\ \Omega$; $I = E_{eq}/R_{total} = (5/3)/2 = 5/6$ A.

The parallel cells reduce to Eeq = 5/3 V with req = 2/3 Ω; adding the 4/3 Ω load gives 2 Ω total, so I = $(5/3)/2 = 5/6$ A.

Q34.

Answer: (c) $\rho' = \rho$, $R' = 16R$, $P' = (1/16)P$

- **Geometry after stretching:** Radius halved → $A' = \pi(r/2)^2 = A/4$; constant volume → $L' = 4L$.
- **New resistance:** $R' = \rho L'/A' = \rho(4L)/(A/4) = 16\rho L/A = 16R$; resistivity is a material property, $\rho' = \rho$.
- **New power rating:** $P' = V^2/R' = V^2/(16R) = (1/16)P$.

Halving the radius quarters the area and (at constant volume) quadruples the length, so R grows 16×; resistivity is unchanged and at the same voltage $P = V^2/R$ falls to $P/16$.

Q35.

Answer: (c) (B), (A), (C)

- **Susceptibility by class:** Diamagnetics: very small negative susceptibility (weakly repelled; e.g. bismuth, copper). Paramagnetics: small positive susceptibility (weakly attracted; e.g. aluminum, platinum). Ferromagnetics: very large positive susceptibility (strongly attracted; e.g. iron, nickel, cobalt).
- **Order:** Increasing susceptibility: Diamagnetics (B) < Paramagnetics (A) < Ferromagnetics (C).

Susceptibility runs from slightly negative (diamagnets) through slightly positive (paramagnets) to very large positive (ferromagnets), giving order B, A, C.

Q36.

Answer: (c) proportional to $I_1 \times I_2 \times L$

- **Ampère's force law:** $F/L = (\mu_0/2\pi)(I_1 I_2)/d$ between two parallel conductors.
- **Force on length L:** $F = (\mu_0/2\pi)(I_1 I_2 L)/d$ — proportional to both the product of the currents and the length.

The force per unit length between parallel currents is $(\mu_0/2\pi)I_1 I_2/d$, so the force on a stretch L scales with

$I_1 \cdot I_2 \cdot L$.

Q37.

Answer: (a) $(\mu_0 I)/(4r)$ out of the plane

- **Current division:** The 3I current divides inversely with resistance: I through ABC (2R) and 2I through ADC (R).
- **Field of each semicircle:** $B_{ABC} = \frac{1}{2} \cdot \frac{\mu_0 I}{2r} = \frac{\mu_0 I}{4r}$; $B_{ADC} = \frac{1}{2} \cdot \frac{\mu_0 2I}{2r} = \frac{2\mu_0 I}{4r}$.
- **Net field:** The fields oppose: $B_{total} = B_{ADC} - B_{ABC} = \frac{\mu_0 I}{4r}$; by the right-hand rule the net field is out of the plane.

The branch currents (I and 2I) create opposing semicircular fields μ_0I/4r and 2μ_0I/4r at the center; the difference μ_0I/4r points out of the plane by the right-hand rule.

Q38.

Answer: (b) 2×10^{-4} Nm

- **Torque formula:** $\tau = nIAB \sin\theta$ with n = 1, I = 10 A, $A = 1\ \text{cm}^2 = 1 \times 10^{-4}$ m^2, B = 0.2 T.
- **Angle:** Field parallel to the loop's plane $\rightarrow$ angle between the normal and the field is 90°, so $\sin\theta = 1$.
- **Evaluate:** $\tau = 1 \times 10 \times 10^{-4} \times 0.2 \times 1 = 2 \times 10^{-4}$ Nm.

With B in the plane of the loop the torque is maximal: τ = IAB = 10 $\times$ 10^{-4} $\times$ 0.2 = 2 $\times$ 10^{-4} Nm.

Q39.

Answer: (d) purely capacitive

- **Phase relations:** Purely resistive: current and voltage in phase. Purely inductive: current lags voltage by $\pi/2$. Purely capacitive: current leads voltage by $\pi/2$.
- **Match:** Since the current leads the voltage by $\pi/2$, the circuit must be purely capacitive.

A 90° current lead is the signature of a pure capacitor; inductors make current lag and resistors keep it in phase.

Q40.

Answer: (c) 1.5 H

- **Definition:** $M = \Delta\Phi/\Delta I$ — the ratio of flux change in one coil to the current change causing it.
- **Substitute:** $M = 15$ Wb/10 A $= 1.5$ H.

Mutual inductance is flux change per unit current change: 15 Wb / 10 A = 1.5 H (the 0.25 s is not needed).

Q41.

Answer: (b) anticlockwise in both (a) and (b)

- **Figure (a):** Field into the plane; reshaping the irregular wire into a circle increases the enclosed area, increasing the inward flux. To oppose this increase, the induced current flows anticlockwise (producing a field out of the plane).
- **Figure (b):** Field out of the plane; collapsing the circular loop into a straight wire decreases the enclosed area, decreasing the outward flux. To oppose this decrease, the induced current flows anticlockwise.
- **Source note:** The printed explanation's final sentence misstates 'clockwise in (a) and anticlockwise in (b)', contradicting its own body; the Answer Key marks (b): anticlockwise in both.

By Lenz's law, (a) opposes growing inward flux with an outward-field (anticlockwise) current, and (b) opposes shrinking outward flux by sustaining it — also anticlockwise.

Q42.

Answer: (b) (A) – (IV), (B) – (III), (C) – (II), (D) – (I)

- **Constant graph:** Resistance (IV) remains constant with frequency → the horizontal line, graph (A).
- **Rising line:** Inductive reactance (III), $X_L = 2\pi f L$, increases linearly with frequency → straight line with positive slope, graph (B).
- **Falling curve:** Capacitive reactance (II), $X_C = 1/(2\pi f C)$, decreases hyperbolically with frequency → graph (C).
- **Dip curve:** Impedance (I) combines resistance and reactance without a simple linear/hyperbolic form (dips near resonance) → graph (D). (The printed final matching line misprints (B)–(II), (C)–(III); the key gives option (b).)

Resistance is frequency-independent (flat), inductive reactance grows linearly, capacitive reactance falls as 1/f, and impedance shows the dipped curve — matching (A)-(IV), (B)-(III), (C)-(II), (D)-(I).

Q43.

Answer: (a) 1 : 1

- **Stated result:** The ratio of energy densities of the electric and magnetic fields in an electromagnetic wave is 1 : 1.
- **Reason:** The wave carries its energy equally in the electric and magnetic fields ($\frac{1}{2}\varepsilon_0 E^2 = \frac{B^2}{2\mu_0}$ on average), so neither field dominates.

In an EM wave the average electric and magnetic energy densities are exactly equal, so the ratio is 1 : 1.

Q44.

Answer: (c) Radio waves, microwaves, Infrared waves, visible waves, X-rays

- **Recall the spectrum:** Across the electromagnetic spectrum, wavelength decreases from radio waves through microwaves, infrared and visible light to X-rays.
- **Pick the matching option:** From the given options, the correct arrangement in decreasing order of wavelength is: Radio waves, microwaves, Infrared waves, visible waves, X-rays.

Wavelength shrinks along the spectrum from radio (longest) through microwaves, infrared and visible light down to X-rays (shortest of the listed).

Q45.

Answer: (b) (A) – (II), (B) – (III), (C) – (IV), (D) – (I)

- **Microwaves and infrared:** Microwaves are commonly produced using a magnetron (electrons interacting with a magnetic field); infrared waves are produced by the vibration and rotation of atoms and molecules.
- **X-rays and radio waves:** X-rays are produced by bombarding a large atomic number metal target with fast-moving electrons; radio waves are commonly generated using an LC (inductor–capacitor) oscillator.

Each wave has a standard production route: magnetron → microwaves, molecular vibration → infrared, electron bombardment of heavy metal targets → X-rays, LC oscillator → radio waves.

Q46.

Answer: (a) 16 cm left of P in air

- **Set up refraction formula:** $\dfrac{\mu_2}{v} - \dfrac{\mu_1}{u} = \dfrac{\mu_2 - \mu_1}{R}$ with $\mu_1 = 1$, $\mu_2 = 4/3$, $u = -20$ cm, $R = -10$ cm.
- **Substitute:** $\dfrac{4/3}{v} - \dfrac{1}{-20} = \dfrac{4/3 - 1}{-10}$.
- **Solve:** $v = -16$ cm; the negative value indicates the image is formed to the left of the surface — 16 cm left of P, in air.

Refraction at the single spherical surface gives v = -16 cm; a negative image distance places the (virtual) image 16 cm to the left of P, on the air side.

Q47.

Answer: (a) $P \propto (\mu - 1)$

- **Lens maker's formula:** $P = (\mu - 1)\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right)$, where μ is the refractive index and R_1, R_2 the radii of curvature.
- **Fixed radii:** With R_1 and R_2 fixed, $P \propto (\mu - 1)$.

By the lens maker's formula, with the curvature factor fixed the power scales directly with (μ - 1).

Q48.

Answer: (b) Graph (b): hyperbolic branch in the second quadrant (negative u, positive v)

- **Lens formula:** $\dfrac{1}{f} = \dfrac{1}{v} - \dfrac{1}{u}$, rearranged to $v = \dfrac{uf}{u - f}$.
- **Shape and quadrant:** The graph of this function is a hyperbola; it lies in the 2nd quadrant as u is negative (v positive).

With the sign convention u < 0 for a real object and v > 0 for a real image, v = uf/(u-f) traces a hyperbola in the second quadrant — graph (b).

Q49.

Answer: (c) 5×10^{-7} m

- **Central-maximum width:** $\beta = \dfrac{2\lambda D}{a}$, so $\lambda = \dfrac{\beta a}{2D}$; given a = 0.1 mm, β = 5 mm, D = 0.5 m.
- **Substitute and evaluate:** $\lambda = \dfrac{(5 \times 10^{-3})(0.1 \times 10^{-3})}{2 \times 0.5} = 5 \times 10^{-7}$ m.

The single-slit central maximum has width 2λD/a; inverting with the measured 5 mm width gives λ = 5 × 10^{-7} m (500 nm).

Q50.

Answer: (c) Maximum kinetic energy of photoelectrons emitted can be $h\nu_0$

- **Photon and threshold energies:** Incident photon energy $E = h(2\nu_0)$; threshold energy $E_0 = h\nu_0$.
- **Einstein's equation:** The excess energy becomes kinetic energy: $K.E._{max} = h(2\nu_0) - h\nu_0 = h\nu_0$.

By Einstein's photoelectric equation the maximum kinetic energy is hν - $h\nu_0$ = $h(2\nu_0)$ - $h\nu_0$ = $h\nu_0$; emitted electrons can have up to (not all exactly) this energy.